

Original Research

Explainable variational autoencoder (E-VAE) model using genome-wide SNPs to predict dementia

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ABSTRACT

Objective: Alzheimer's disease (AD) and AD related dementias (ADRD) are complex multifactorial neurodegenerative diseases. The associations between genetic variants obtained from genome wide association studies (GWAS) are the most widely available and well documented variants associated with ADRD. Application of deep learning methods to analyze large scale GWAS data may be a powerful approach to elucidate the biological mechanisms in ADRD compared to penalized regression models that may lead to over-fitting.

Methods: We developed a deep learning framework explainable variational autoencoder (E-VAE) classifier model using genotype (GWAS SNPs = 5474) data from 2714 study participants in the Health and Retirement Study (HRS) to classify ADRD. We validated the generalizability of this model among 234 participants in the Religious Orders Study and Memory and Aging Project (ROSMAP). Utilizing a linear decoder approach we have extracted the weights associated with latent features for biological interpretation.

Results: We obtained a predictive accuracy of 0.71 (95 % CI [0.59, 0.84]) with an AUC of 0.69 in the HRS test dataset and got an accuracy of 0.62 (95 % CI [0.56, 0.68]) with an AUC of 0.63 in the ROSMAP dataset.

Conclusion: This is the first study showing the generalizability of a deep learning prediction model for dementia using genetic variants in an independent cohort. The latent features identified using E-VAE can help us understand the biology of AD/ ADRD and better characterize disease status.

1. Introduction

1.1. Statement of significance

Problem Lack of generalizable and biologically interpretable classifier for dementia using genetic variants.

What is Already Known Previous studies have developed prediction models for AD/ ADRD using Polygenic Risk Scores or relevant SNPs (n = 50–1000) from GWAS. These studies failed to show the generalizability of the model in external independent dataset.

What This Paper Adds We developed a framework using deep neural network model (E-VAE classifier) for dementia classification that accounts for the higher order interactions of input features. We demonstrated the biological interpretability of the model using a linear decoder and demonstrated the generalizability of the model in an independent cohort.

Alzheimer's disease (AD) and AD related dementia (ADRD) are complex multifactorial neurodegenerative disorders [14,26], characterized by memory loss and cognitive decline affecting a person's activities of daily living [12]. Studies have shown that dysfunctional interactions of genes and biochemical changes occur many years before the onset of clinical symptoms of AD/ADRD [13,14,26]. AD/ADRD doesn't have a clear inheritance pattern and studies have shown that risk of dementia is polygenic, in that it involves multiple genes with minor effects or the interaction between genes [2]. Identification of susceptibility genes for complex and heterogeneous multi-factorial human diseases has been a significant challenge in human genetics [31]. In the last couple of decades, numerous GWAS studies (146 studies in GWAS catalog) along with large scale consortia powered by meta-analyses of GWAS studies have identified thousands of common genetic variants associated with AD/ ADRD [10]. A large study among Swedish twins also showed the

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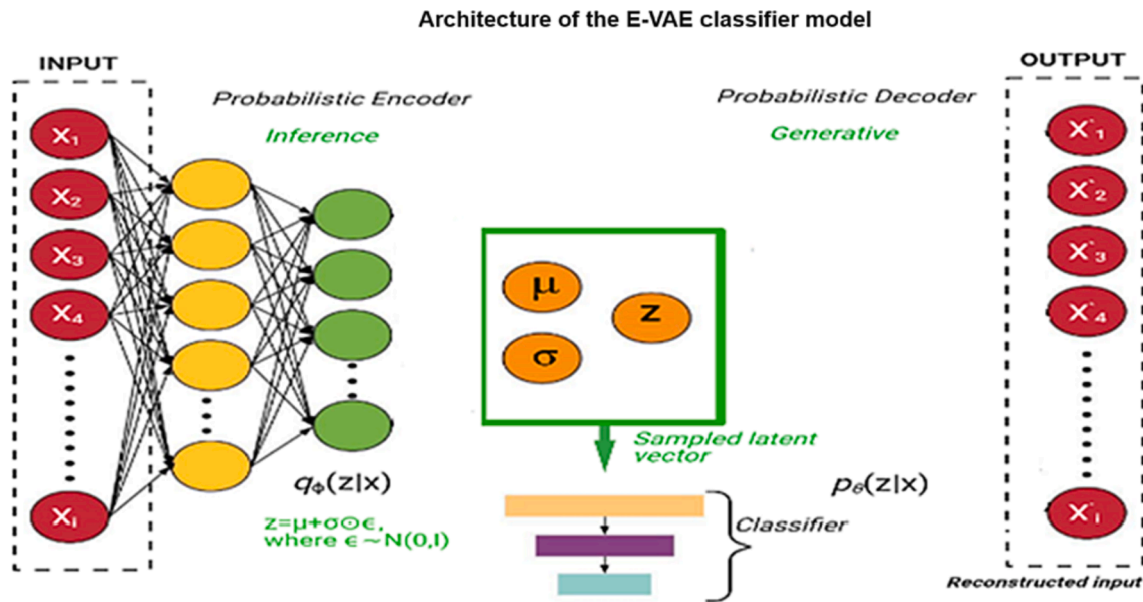


Fig. 1. Architecture of the E-VAE classifier model: The diagram showing the basic architecture of the unsupervised phase of VAE with a fully connected encoder layer and a linear decoder and supervised phase using a Classifier network to predict dementia. Each node and edges represent a fully connected network. X_i represents each input feature to the inference network and X'_i is the reconstructed input feature from the generative network. The sampled latent vector 'z' with the specified latent dimension was then used as input for the classifier network to classify participants in the study.

heritability of AD among older adults to be 58 % [7]. This explosion of information in human genetics and epidemiology have created new challenges for modeling and interpretation. Challenges in utilizing genome-wide approach for disease classification include the failure of traditional statistical models to account for the higher order interaction in the genome-wide approach [31] and the challenge in selecting useful features for disease predictions from a list of candidate features. Previous studies have developed prediction models for AD/ADRD using Polygenic Risk Scores (based on genome-wide significant loci of AD) or by using relevant Single Nucleotide Polymorphisms (SNPs) ($n = 50\text{--}1000$) from GWAS data [5,24,32,37]. With advancement in newer machine learning (ML) methods, feature selection algorithms have gained a lot of attention. Feature selection in ML is important to develop optimized algorithms to improve target prediction [19]. To address these challenges, we developed a framework using a complex deep neural network model that allows for a large list of features as inputs, accounts for the higher order interactions of features using non-linear model fitting, and enables the interpretation of the importance of features in the context of disease biology.

A study using Disentangled Sequential Graph Autoencoder which utilize the Sequential Variational Autoencoder (SVAE), graph convolution framework to understand the disentangled representation of the measurements over the entire regions of interest to characterize the brain by considering the time-variant and time-invariant latent variables [36]. Recently, a study using ADNI genotype data utilized a deep learning-based Convolutional Neural Network (CNN) approach for SNPs selection and developed a classification model for dementia that achieved an area under the curve (AUC) of 0.82 (Jo, Nho, Bice, Saykin, & For The Alzheimer's Disease Neuroimaging, 2022). Using more advanced deep autoencoder or variational autoencoder algorithm allow us to learn compressed feature representations from high dimensional genome data. In this study, we utilized a deep generative neural network prediction model with a basic structure of Variational Autoencoder (VAE) and a classifier network for supervised prediction. VAEs are a particular type of deep generative model that combines a Bayesian paradigm with a neural network model that can approximate complex, non-linear transformations to learn the latent representation of features from input omics data using variational inference [17,18]. Thus, unlike

the traditional autoencoders, VAEs can handle higher order interactions between high dimensional input features and enhance the classification performance.

In this paper, we developed a novel end to end deep learning based and explainable variational autoencoder (E-VAE) classifier model, that includes an unsupervised phase using a VAE that generate a latent embedding of the input features and a classifier network that use the learned latent representations to classify a binary outcome. We implemented the explainable model by utilizing a linear decoder layer to extract the learned weights for input features from the trained model. To develop and validate our E-VAE classifier model, we used genotype data from the Health and Retirement Study (HRS) as input features to predict dementia. We used an association-based feature selection method for dimension reduction and a balanced-class outcome to improve the sensitivity of the model. We demonstrated the generalizability of the E-VAE classifier model in an external dataset from the Religious Orders Study and Rush Memory and Aging Project (ROSMAP) and interpreted the biological functions represented by the latent features from the E-VAE classifier model. To the best of our knowledge this is the first study utilizing a VAE model to use genetic variants to classify dementia and interpret the biological importance of low dimensional latent variable embedding.

2. Materials and methods

2.1. Explainable variational autoencoder (E-VAE) classifier model and its optimization

The E-VAE classifier model is an end to end deep learning model with an unsupervised phase of VAE and a supervised phase of classifier network. Variational autoencoder is a neural network that consists of an encoder, bottleneck layer and a decoder. Compared to classical autoencoders, VAE has a more regularized encodings distribution to avoid overfitting and the specific properties of latent space enable new sample generation/data reconstruction [18]. We adapted the basic architecture of OmiVAE with a fully connected encoder network and classifier network [38] to classify participants with dementia and we have implemented a linear decoder to extract the weights associated

with input features and latent variable embedding.

The Fig. 1 illustrates the basic architecture of E-VAE classifier model.

We considered the GWAS SNPs dataset ‘D’ with ‘N’ samples and ‘m’ omics molecular features, and assumed that each sample $x^i \in \mathbb{R}^m$ is generated using latent vector $z^i \in \mathbb{R}^p$, where $p \ll m$. In this process, each latent variable z^i is generated from prior distribution: $p_\theta(z)$ and each sample x^i is generated from the conditional distribution: $p_\phi(x|z)$; and the posterior probability distribution is $p_\theta(z|x)$. VAE use variational Inference (VI), it is a technique of inference in the graphical model, with x the input variable and z the hidden variable. We had to compute the $p(z|x) = \frac{p(x|z)p(z)}{p(x)}$, but the marginal distribution $p(x) = \int p_\theta(x|z)p_\theta(z)dz$ is complex to compute as it is intractable especially in the high dimensional space. So, we approximate a feasible Gaussian distribution $q_\phi(z|x)$ in the probabilistic encoder using VI. Here θ and ϕ are learnable parameters/ weights of the decoder (generative) and encoder (inference) networks respectively.

The loss function of VAE has two terms, the first term to penalize reconstruction error with the objective function to maximize the reconstruction likelihood of input data and the second term to minimize KL divergence regularization with the objective to have the learned distribution $q_\phi(z|x)$ to be similar to the true posterior distribution $p_\theta(z|x)$.

The final objective function of VAE is in equation (1)

$$\text{argmin}_{\theta, \phi} \mathcal{L}_{\text{VAE}}(\theta, \phi; x) = \mathcal{L}_{\text{RECONS}}(\theta, \phi) + \mathcal{L}_{\text{KLD}}(\theta, \phi) \quad (1)$$

$$= \mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)] - D_{\text{KL}}[q_\phi(z|x) || p_\theta(z|x)]; \text{ where } \mathbb{E}[\cdot] \text{ is mathematical expectation and } D_{\text{KL}}(\cdot || \cdot) \text{ is the KL-divergence term.}$$

In the VAE framework, the encoder and decoder network were jointly optimized by maximizing the evidence lower bound (ELBO) $\log p_\theta(x) \geq \mathcal{L}_{\text{VAE}}(\theta, \phi; x)$. We have minimized the loss in order to maximize the lower bound of the probability of generating real data samples as shown in the equation (2)

$$\log p_\theta(x) \geq \mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)] - D_{\text{KL}}[q_\phi(z|x) || p_\theta(z|x)] \quad (2)$$

In the unsupervised phase of E-VAE classifier model, the encoder layer encodes a Gaussian distribution $N(\mu, \sigma)$ with a mean (μ) and a standard deviation (σ) for input sample x_i over the latent space. Then the latent representation z was sampled from the Gaussian distribution $q_\phi(z|x)$. The decoder layer will reconstruct the input sample (x_i) using latent representations (z) and the VAE loss function will optimize the algorithm.

Reparameterization trick: Sampling from $z \sim q_\phi(z|x)$ is a stochastic process, so we cannot directly backpropagate the gradient. We used the reparameterization trick to use a stochastic gradient to train the VAE network, by multiplying the variance of hidden distribution with a random variable (ϵ) sampled from unit normal distribution $N(0,1)$ and back propagate to identify the parameter of encoder. Then the whole network is differentiable and we can use optimization techniques to solve inference problem successfully. The distribution $q_\phi(z|x)$ is a multivariate Gaussian with a diagonal covariance structure estimated as shown in the equation (3)

$$z \sim q_\phi(z|x) = \mu + \sigma \odot \epsilon, \text{ where } \epsilon \sim N(0, I) \quad (3)$$

The classifier network: The VAE encoder layer learned the latent representations in the unsupervised phase and in the supervised phase the latent variables were used as input to the classifier network to predict dementia. The E-VAE classifier loss was estimated using the binary cross-entropy loss as in equation (4) where ‘y’ was the observed class label (Dementia or Normal) and ‘p’ was the predicted probability of that class label.

$$\mathcal{L}_{\text{Dementia}} = -1/N \sum_{i=1}^N ((y_i \cdot \log(p(y_i)) + (1 - y_i) \cdot \log(1 - p(y_i))) \quad (4)$$

We measured the overall training error using the TOTAL LOSS

function of the VAE classification network by combining the VAE ($\mathcal{L}_{\text{VAE}}$) loss and the classification loss ($\mathcal{L}_{\text{Dementia}}$) as in the equation (5) where α and β are the parameters that weight the loss term during training.

$$\mathcal{L}_{\text{TOTAL}} = \alpha \mathcal{L}_{\text{VAE}} + \beta \mathcal{L}_{\text{Dementia}} \quad (5)$$

The classification network imposed additional regularization for the classification task and learned latent representations by accurately reconstructing the input and to classify the participants with dementia from normal. We used the Adam optimizer to train the E-VAE classifier model and used Rectified Linear Units (ReLU) activation function and batch normalization to each fully connected block in the encoder hidden layers, a sigmoid activation was used in the linear decoder and a softmax function employed in the final layer of classifier.

Hyper parameter tuning: In machine learning, especially when we use deep learning models such as VAE, parameter tuning is a key part of fitting models to data. We tuned the hyper parameters such as epoch, batch size, number of hidden layers, number of nodes in each layer and number of latent features and learning rate using Optuna; a hyper-parameter optimization framework. Optuna is a python library that enables us to tune the machine learning model automatically. We optimized the hyperparameters to maximize the accuracy for prediction using HRS validation dataset along with an early stopping rule for training the model.

Model evaluation metrics: We evaluated the performance of GWAS-based E-VAE classifier model based on the total accuracy score (fraction of correct predictions) and calculated sensitivity and specificity at the cutoff showing best accuracy. We also used a Receiver-operating characteristic (ROC) analysis to evaluate the non-random prediction performance of the models and used the area under the curve (AUC) to compare the performance of various prediction models [39].

VAE model Implementation: The E-VAE classifier model was built in Python (version 3.8) with PyTorch module (version 1.7.0). We used GPU cluster in Minnesota Supercomputer Institute (MSI) Mesabi system for a faster implementation of the GWAS-based VAE classifier model.

2.2. Comparison of VAE model with penalized logistic regression (LR) models

We used penalized logistic regression model, an ElasticNet model, to compare the performance of the E-VAE classifier model. We optimized the penalized LR model by tuning the hyper parameters such as penalty type and inverse of regularization strength.

2.3. Biological interpretation of latent embedding

We adapted the method described in Tybalt VAE [34] model to extract weights from VAE decoder layer to learn the biological signals embedded in the latent features. Due to the complex architecture of encoder layer, we extracted the weights using a linear decoder that captured the contribution of each input SNPs to the learned latent features. We did z-score normalization of the weights and evaluated the distribution of weights of all encoded features to select the positive high and negative high weighted SNPs that were 2 SD above or below the mean weight. We used geneprofiler2 to convert SNPs to the gene loci. We utilized the WEB-based Gene Set Analysis Toolkit (WebGestalt) to characterize the top weighted genes from the E-VAE classifier model and performed over representation analysis to translate the top weighted gene list into biological insights.

2.4. Data

We utilized the genotype data along with cognitive function scores measured in the HRS 2016 survey to build the E-VAE classifier model. The HRS is a biennial survey of nationally representative sample of older adults aged 51 years and older. Since 1992, HRS includes measures of cognitive function to understand the onset and impact of cognitive

Table 1

Distribution of participant characteristics in the study cohort.

Variables	HRS (N = 2714)	ROSMAP (N = 234)
Age (years)	69.7 ± 9	86.9 ± 4.7
Sex (% females)	1594 (58.7 %)	170 (72.6 %)
Education (years)	13.4 ± 5.5	15.6 ± 3.1
Dementia (%)	128 (4.7 %)	12 (5.1 %)

impairment using an in person and telephone surveys [4]. The Table 1 shows the distribution of participant characteristics in the HRS 2016 survey who participated in the Harmonized Cognitive Assessment and Protocol (HCAP) Study. We also obtained genetic variants data from the Religious Orders Study (ROS) and Rush Memory and Aging Project (MAP), (ROSMAP) from the AD knowledge Portal [9] and accessed the cognitive function measures and basic demographics from RUSH Alzheimer's Data Center to evaluate the generalizability of the E-VAE classifier in an independent dataset. The ROSMAP studies are longitudinal epidemiological and clinical-pathological cohort study initiated in 1994 and 1997 respectively. The Table 1 shows the distribution of participant characteristics in the ROSMAP who had cognitive function measures and genotype data. The ROSMAP study prospectively collected risk factors of cognitive impairment and measured AD dementia incidence among nuns/ brothers, priests and laypersons along with a structured neuropathological examination after death [3].

2.5. Ascertainment of dementia in the HRS

Cognitive performance tests in the 2016 HRS survey included measures of episodic memory (using immediate and delayed 10-noun free recall test) and measures of mental processing and working memory (counting backward from 20, serial 7 subtraction). These scores were combined to create a composite score ranging from 0 to 27 that was used to estimate the cognitive function. We used the Langa-Weir (LW) classification approach using the 27 points cognitive function score to group participants into two categories: Dementia (0–6); and Normal (12–27) [4]. Among 2714 participants in the HRS, 128 (4.7 %) participants had dementia and 2586 (95.3 %) participants had normal cognition in the 2016 survey.

2.6. Ascertainment of dementia in the ROSMAP study

A clinical diagnosis of cognitive status was rendered based on a three-stage process including a battery of 19 cognitive tests (comprising 5 domains i.e. episodic memory, visuospatial ability/perceptual orientation, perceptual speed, semantic memory and working memory), a clinical review by a neuropsychologist and a diagnostic classification by a clinician. Clinical diagnosis of AD was based on criteria of the joint working group of the National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's disease and Related Disorders Association (NINCDS/ADRDA). Individuals without dementia or mild cognitive impairment (MCI) are categorized as having no cognitive impairment (NCI). Among 234 participants in the ROSMAP study 12 (5.1 %) participants had dementia and 222 (94.9 %) participants had normal cognition.

2.7. Harmonization of dementia definition/cognition measures in HRS and ROSMAP study

To optimally harmonize cognition domain scores measured in HRS and ROSMAP using commonly available cognition tests between the two studies, we used z-scores estimated. In the HRS, we used the immediate word recall, delayed word recall, serial 7's test and in the ROSMAP study, we used East Boston immediate recall, East Boston delayed recall and digits backward which were used to measure working memory and episodic memory. We estimated the cumulative cognition score from the

three domains and then used the z-score normalization to harmonize the cognition score. We have identified the z-score cut off as '-1.53' in HRS at which there will be no discrepancy with the dementia classification based on published LW classification approach. Similar to the LW classification approach in HRS [4], we classified participants as Normal and Dementia. We used same z-score cut off to classify participants in ROSMAP and compared the classification accuracy using the diagnosis category reported in ROSMAP based on over all cognitive function, neuropsychologist and clinician review. We removed CIND/MCI participants from both studies to keep the outcome class binary. We also removed 7 participants who were classified as having dementia in the ROSMAP study but had HRS z-score based cognitive scores in the normal range.

2.8. HRS training and validation data with Dementia:Normal class balance

We have included 2714 (Dementia = 128, Normal = 2586) participants from the HRS study. As we had only ~ 5 % prevalence of dementia, we randomly selected 128 participants from the 'Normal' class and the 128 participants with dementia to balance the class distribution to train the E-VAE classifier model. We used a five-fold stratified split to select training and validation data. We combined the first four folds for training data (N = 204) and used the fifth fold as validation data (N = 52) with equal number of participants in dementia and normal class.

2.9. HRS GWAS data

The HRS genotyped salivary DNA samples collected since 2006 at multiple time points during field visits yielding a total of 18,916 unique participants. We used the most recent version of genotype data from 2006 to 2012. The genotyping was performed by NIH Center for Inherited Disease Research using the Illumina HumanOmni2.5-4v1/8v1 array and genotyping QC analysis was performed at the University of Michigan using HumanOmni2.5-4v1 H for SNP annotation. The Michigan Imputation Server (<https://imputationserver.sph.umich.edu/>) was used for imputation to the 1000 Genomes Project phase 3 integrated variant set (v5, released Oct 2014). The QC filters that were applied to the genotype data prior to imputation included SNP Hardy-Weinberg Equilibrium (HWE) p value < 10⁻⁴, SNP missing call rate ≥ 2 %, and other criteria as specified in Table 1 of the HRS "Quality Control Report for Genotypic Data" [28]. As stated in the chapter by Moore [25], we used a filter approach for feature selection in order to reduce the computational limitation using millions of SNPs. We selected genome-wide significant SNPs (p < 5e-08) curated in HRS based on already published large-scale GWAS studies related to ADRD (Gencog:11598 SNPs [6], IGAP_AD:576 SNPs [21,20], Kunkle_AD_wAPOE:1269 SNPs [21,20], Wightman_AD:3570 SNPs [35]). We filtered out poor quality and rare SNPs as part of the feature selection. We filtered out SNPs that have low imputation info score < 0.9 and got 14,662 unique SNPs from these studies.

2.10. ROSMAP study GWAS data

To validate the generalizability of the GWAS-based E-VAE classifier model for dementia trained using genetic variants data in the HRS, we utilized the genotype data obtained from blood along with clinical and syndromic phenotype data in the ROSMAP study. In ROSMAP, we had imputed dosage data from two batches of genotype runs; Illumina chop samples (N = 382) and Affymetrix (N = 1708) samples with 37 million variants. Based on sample-level quality control assessment, excluded samples with genotype success rate < 95 % and based on SNP-level quality control assessment excluded SNPs with HWE p-value < 0.001, MAF < 0.01, genotype call rate < 0.95, etc. as described in AD Knowledge portal. Population outliers were identified and removed using EIGENSTRAT with default parameters.

Table 2

Comparison of prediction performances between the VAE classifier model and penalized logistic regression model in the HRS and ROSMAP study.

Models	Accuracy	Sensitivity/ Recall	Specificity	AUC
E-VAE classifier model using HRS test (N = 52)	0.71	0.73	0.69	0.69
E-VAE classifier model using ROSMAP test (N = 234)	0.62	0.58	0.62	0.63
Penalized LR model HRS test (N = 52)	0.69	0.77	0.62	0.72
Penalized LR model ROSMAP test (N = 234)	0.80	0.17	0.83	0.43

2.11. Harmonization of GWAS SNPs in the HRS and ROSMAP study

Both studies employed similar SNP-level quality control assessment. We used the unique list of GWAS SNPs in the HRS and matched SNPs from the ROSMAP study genotype data and identified 13,507 SNPs in common. We have filtered out SNPs that were missing in at least one participant in HRS or ROSMAP participants and finalized 5474 SNPs as input features for the E-VAE classifier model.

3. Results

3.1. Training and validation of the E-VAE classifier model in the HRS

The 204 participants in the training set included 102 dementia cases, 110 women with an average age of 73 years. The 52 participants in the validation set included 26 dementia cases, 34 women, with an average age of 72 years. Table 2 shows the comparison of E-VAE classifier model performance with a penalized logistic regression (LR) model using 5474 SNPs in the HRS validation data and the Fig. 2 shows the AUC-ROC

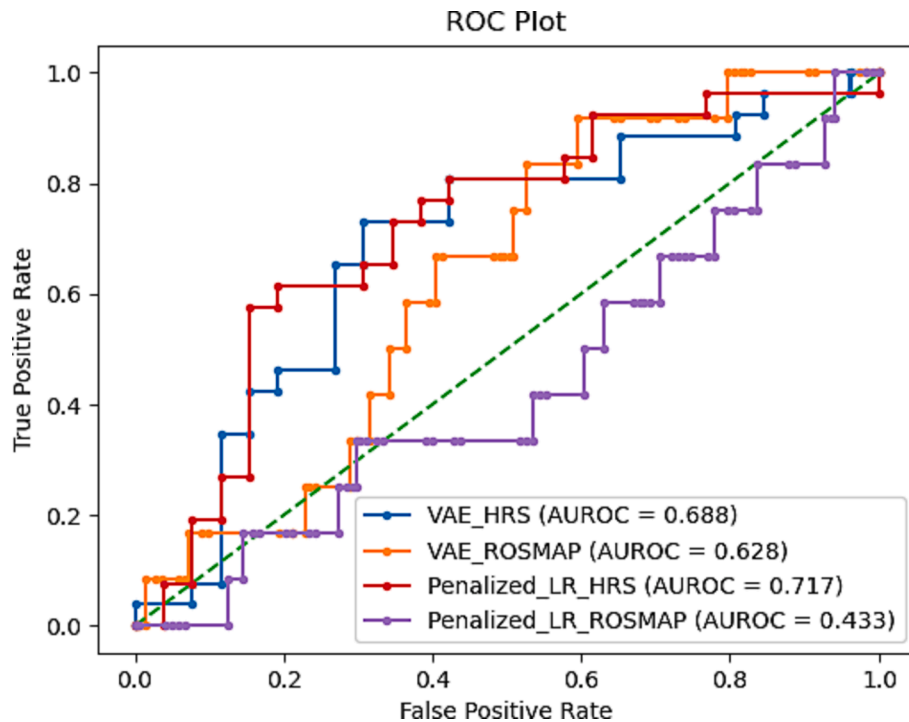


Fig. 2. Comparison of AUC-ROC between the E-VAE classifier model and penalized logistic regression model in the HRS and ROSMAP study.

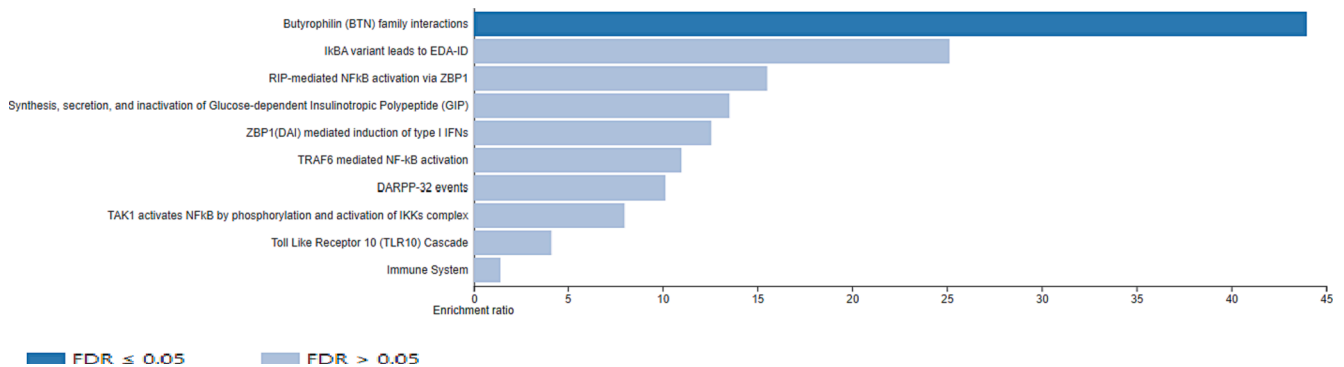


Fig. 3. Results of over representation analysis (ORA) using Web based gene set analysis toolkit (WebGestalt). This figure shows the pathway enrichment results of ORA using Reactome. We used 120 unique entrezgene IDs that have functional categories annotated for enrichment analysis. We used the Benjamini-Hochberg (BH) method adjusted FDR p-value of 0.05 to select over enriched pathways.

curve. The E-VAE classifier model achieved an accuracy of 0.71 with 95 % CI [0.59, 0.84] and sensitivity of 0.73 in the HRS validation data. We found that the E-VAE classifier had higher predictive accuracy but lower sensitivity in the HRS validation data compared the penalized LR model accuracy of 0.69 and sensitivity of 0.77.

3.2. Generalizability of the E-VAE classifier model in the ROSMAP study

The 234 participants in the ROSMAP study included 12 (5 %) dementia cases, 170 women, with average age of 84 years. The parameters of the E-VAE classifier model with highest accuracy in HRS validation data was fitted to the ROSMAP data to evaluate the generalizability of the model performance in an independent dataset. The E-VAE classifier model got an accuracy of 0.62 with 95 % CI [0.56, 0.68] with an AUC of 0.63 in the ROSMAP data compared to the accuracy of 0.80 with an AUC of 0.43 using the penalized LR model. We found that the E-VAE classifier model had higher generalizability in external dataset compared to traditional LR model. The E-VAE classifier model had better sensitivity to classify dementia in the ROSMAP study, whereas the penalized logistic regression model showed poor generalizability in ROSMAP data.

3.3. Biological interpretability of the latent embedding of GWAS SNPs

As shown in Fig. 3, the ORA using Reactome pathways we found that genes in *Butyrophilin (BTN)* family interactions were over enriched in the list of top weighted genes from the E-VAE classifier model. We extracted top weighted 264 genes based on z-score cut off of 2 and identified 120 unique entrezgene IDs with functional categories annotated. Among these genes used in the over representation analysis, 6 genes were in the *Butyrophilin (BTN)* family interactions (gene set of 12 genes) in Reactome (<https://reactome.org/PathwayBrowser/#/R-HSA-8851680>). We did not observe any pathways enriched using KEGG pathways in the ORA analysis.

4. Discussion

In this study, we used an end-to-end deep learning (DL) model with the basic structure of VAE, a probabilistic Autoencoder (AE) model [18], to extract low dimensional feature representations from genetic variants and utilize a classifier network to predict dementia [38]. The GWAS-based E-VAE classifier model achieved an accuracy of 0.71 with an AUC of 0.69 in the HRS test data and achieved a generalizable accuracy of 0.62 with an AUC of 0.63 in the ROSMAP study data. We found that the E-VAE classifier model had better generalizability in external dataset compared to a penalized logistic regression model. We demonstrated the utility of a linear decoder to extract the input feature weights for the latent embedding of SNPs for the biological interpretability by optimizing a classifier model for dementia. We found that genes from *Butyrophilin (BTN)* family interactions gene sets were over-represented in Reactome pathways.

Genome-wide association studies (GWAS) of AD have identified many new genetic loci [15,22], affecting pathways of immunity, lipid metabolism, tau binding proteins, glucose metabolism, mitochondrial function [23] and amyloid precursor protein (APP) metabolism [8,21,20]. However, a genome-wide complex trait analysis showed that all known genetic variants can collectively explain only 15 % of the variance in AD even though heritability of AD has been estimated to be 58 % in twin studies [7,30]. Generally, predicting neurodegenerative disorders are challenging as the AD pathological information are not easily accessible [11]. To address this issue, developing computational methods to identify novel gene candidates from saliva or blood, easily accessible biospecimens, is necessary for early detection and targeted intervention of AD/ADRD. A deep learning-based Convolutional neural network (CNN) model developed by Jo et al. using ADNI genotype data showed the power of deep learning approaches using genetic variants to predict AD [16]. They developed a three step approach to identify

phenotype associated SNPs and to develop a classification model. They mainly identified the well-known APOE region as most significant locus for AD [16] using the phenotype information score. Compared to their reported accuracy of 75 % using a CNN model with 4000 SNPs, we obtained a similar accuracy of 71 % with 5474 SNPs in the HRS test data. However, the previous study did not evaluate the validity of their model in external datasets. We have demonstrated that our findings had a 62 % accuracy in an external dataset. Previous studies have showed that association-based feature selection from the training data can improve prediction performance of ML models [27] and this may partly account for the high accuracy reported with the CNN-based model. In contrast, our study did not select SNPs based on the association in the HRS dataset alone and instead used a list of SNPs that were associated with AD/ADRD in external datasets independent from the datasets used in this analysis. Another advantage of our study is that the GWAS-based E-VAE classifier model we developed is capable of learning a biologically relevant latent space.

A recent study using genotype data compared the predictive performance of different ML models [32] and showed that LASSO model performed the best with SNPs ($n = 1106$) associated with misclassified samples included in the feature pool and achieved an AUC of 0.84. Major limitations of this study were the inclusion of samples from the test dataset for the feature selection and lack of validation of the model performance in an external study. Compared to their feature set, our dataset included much larger number SNPs that could account for the variations in risk of dementia among older adults. We showed that GWAS-based E-VAE classifier model had better generalizability to external dataset compared to traditional penalized logistic regression models. Recently, You et al. developed a dementia prediction model coined as UKB-DRP model using top 10 predictors from 366 candidate variables including genetic, environmental and physical measures collected in longitudinal population based study using UK biobank data achieved an AUC of 0.85 [37]. They used an inner-looped cross validation for hyper parameter tuning and evaluated the model performance in a validation set. The UKB-DRP model implemented using LightGBM algorithm will work only for studies with large sample size and high feature space like UK biobank database and it lack validation of model performance in an external cohort. They identified Age and APOE e4 as major predictors of dementia. We might be able to improve the classification performance of GWAS-based E-VAE classifier model considering APOE and social determinants of health.

Applications of DL-based models in the classification of AD/ADRD phenotypes are still in the nascent stage. Recent advancements in deep learning methods allow us to analyze high throughput data generated from GWAS studies in AD [33] to identify novel biomarkers for early detection. Though GWAS played an important role in identifying the 'candidate genes' or 'candidate loci' susceptible for risk of AD, this information won't be complete if we can't characterize the functional context of these candidate genes in relation to the pathways or other knowledge structures that define the cause-effect relationship [11]. To address this, we have adapted the linear decoder method to extract the decoder weights to identify the contribution the GWAS SNPs to the learned latent features by optimizing a dementia classification network. Among the top weighted genes identified using the GWAS-based E-VAE classifier model, the genes from *Butyrophilin (BTN)* family interactions gene set were over-represented in Reactome pathways. Butyrophilins (BTNs) are a group of the moonlighting proteins in immunoglobulin family. Studies have explored the role of butyrophilin-related molecules in regulate T-cell activation [1] and found to be associated with pathology of many human diseases [29]. The biologically interpretable latent features from the E-VAE classifier model allowed us to explore the complex molecular mechanisms in AD/ADRD etiology and generate new hypothesis to study.

5. Strengths and limitations

A prediction model trained using omics data from one study and validated in an independent study is hard to implement in large cohort studies due to missing data, the technical variation in the measurement of omics data, and differences in the measurement or definition of the outcome variables. In this study, we leveraged results from four large-scale GWAS studies to inform our SNP selection. Since the HRS and ROSMAP employed identical methods for measurement of genetic variants we have implemented a generalizable and explainable E-VAE classifier model for dementia using the GWAS SNPs. The imbalanced datasets severely limit the generalizability of the model as it bias the model to learn to predict the larger class. To overcome this we utilized a balanced-class approach to train the E-VAE classifier model. A major limitation of our study is the small sample size and inability to account for all the heterogeneity observed in genetic studies of AD. Future studies with larger sample sizes can account for the heterogeneity and increase the predictive accuracy of the GWAS-based E-VAE classifier model. We also acknowledge that Polygenic Risk Score (PRS)-based prediction model stands out as a straightforward and highly versatile model when compared to E-VAE classifier model. Our E-VAE classifier model requirement for the presence of all trained SNPs in the test dataset might not align with all study contexts. Another limitation is the definition of dementia used in the HRS and ROSMAP study. The HRS surveys were tailored to identify factors contributing to the development and characterization of cognitive impairment and dementia among older adults of 51 years and older. Compared to the HRS, the ROSMAP study included more targeted population of older adults and captured their cognitive trajectory using a detailed battery of neuropsychological tests and neurological evaluation. We used 3 domains of cognition measures common to both studies and classified participants using a harmonized dementia definition. Also we did not include already known risk factors such as age and APOE allele status in the E-VAE classifier model to predict dementia.

Innovation: Though VAE models have been applied successfully in other health domains such as cancer, the application of VAE model using SNPs in the prediction of AD/ADRD is novel. The biological interpretability of latent features generated from a dementia prediction model is an innovative approach. This allowed us to translate the deep learning-based latent features to characterize the genetic etiology of dementia. In addition, the interpretation of the biological meaning of latent features may lead to new biological hypothesis to be investigated. Availability of genetic variants data from multiple studies of large representative population of older adults allowed for better generalizability of study results.

6. Conclusion

Dementia etiology is complex and is influenced by nonlinear and redundant interactions of genes, when a specific pathway alters from the normal these changes are captured in a genome, epigenome and transcriptome. Utilizing a GWAS-based E-VAE classifier model optimized to predict dementia, we have extracted a biologically relevant latent representations from high dimensional genetic variants data. The E-VAE classifier model accounts for the higher order feature interactions and noise in genome-wide data in the high dimensional space compared to classical dimensionality reduction algorithms. The knowledge gained from this study can be further validated to help develop blood-based signatures for the early identification of AD/ADRD in older adults. Additional functional genomics studies are necessary to identify the genetic pathways to AD/ADRD with the integration of transcriptomic and epigenomic data.

Availability of data

Data used in this manuscript can be accessed from the Health and Retirement Study website (<https://hrsdata.isr.umich.edu/data-products/public-survey-data?ga=2.99940638.1654882771.1674664830-158>

[1526203.1653334037](https://hrsdata.isr.umich.edu/data-products/genetic-data)). The cognition measures and basic demographic data used in this study is publicly available to registered members of the Health and Retirement Study. The genetic variants data can be accessed by authorized investigators and data are distributed through the NCBI Database of Genotypes and Phenotypes (dbGaP), the National Institute on Aging Genetics of Alzheimer's Disease Data Storage Site (NIAGADS), and through the HRS web site (<https://hrs.isr.umich.edu/data-products/genetic-data>). The genetic variants data from the Religious Orders Study (ROS) and Rush Memory and Aging Project (MAP) (ROSMAP) obtained from the AD knowledge Portal using the Synapse account (<https://www.synapse.org/#!Synapse:syn3191087>). We accessed the ROSMAP study cognitive function measures and basic demographics data using a DUA between U of Minnesota and RUSH Alzheimer's Data Center (RADCC).

Ethics approval and consent to participate: All study participants in the HRS and ROSMAP study were consented for study participation and data sharing for research use. These studies were approved by the Institutional Review Board at the University of Michigan, Ann Arbor and the Institutional Review Board of Rush University Medical Center respectively.

CRedit authorship contribution statement

Sithara Vivek: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jessica Faul:** Methodology, Resources, Writing – review & editing. **Bharat Thyagarajan:** Conceptualization, Methodology, Resources, Supervision, Writing – review & editing. **Weihua Guan:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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